

Vision Transformer Driven Mango Leaf Disease Detection with Targeted Spraying Using UAV

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ABSTRACT

Abstract—However, mango (*Mangifera indica*) cultivation in Pakistan is restricted due to the existence of diseases affecting the leaves, including Anthracnose, Powdery Mildew, Bacterial Canker, Dieback, Gall Midge, and Sooty Mould. The conventional method of controlling these diseases includes field inspection and spraying pesticides, which require labor and cost resources, in addition to harming the environment. In this research, we propose an integrated solution, including a leaf disease classifier using Vision Transformer (ViT) architecture and a UAV-based targeted spraying system. We fine-tuned the DeiT-Small model with a customized LayerNorm classification head using a progressive-unfreezing strategy in a stratified split of MangoLeafBD dataset (4,000 images, 8 classes). In order to decrease the domain gap between the training and the application images, an image-to-image translation model using CycleGAN architecture was trained to generate drone-like leaf images. These were added to both training and validation datasets. The accuracy achieved in the validation stage was 97.09%, while the overall accuracy on real and real + synthetic test datasets was 98% with the macro F1-score equal to 0.96. A hexacopter UAV with an imaging system and relay-driven spraying system that would convert disease detection into geo-referenced spraying of pesticides was created based on the literature reporting reductions of 30-50% in pesticide utilization compared to random spraying.

Index Terms—Vision Transformer, DeiT, mango leaf disease detection, CycleGAN, domain adaptation, UAV, precision agriculture, targeted spraying

1. Introduction

Agriculture is still the foundation of many developing countries and mango is considered one of the most economically valuable fruit crops in Pakistan, playing an important role in the economy and exports. Nevertheless, the cultivation of mangoes is greatly limited by various leaf disorders such as Anthracnose, Powdery Mildew, Bacterial Canker, Dieback, Gall Midge, and Sooty Mould. [10].

The majority of disease control programs still employ visual checks, which are costly, laborious, and subject to human error [10], and also involve indiscriminate application of pesticides irrespective of whether there is an infection or not [9]. This not only adds to costs but also makes the field staff and environment vulnerable to chemical exposure.

Deep learning is responsible for automating the recogni-

tion phase. Deep learning models such as CNN, including the ResNet and EfficientNet variants using transfer learning, have achieved classification accuracies of above 99% [6], [13] on standard benchmark datasets of mango leaf diseases. The problem is that CNNs are based on local receptive fields; thus, they fail to learn dependencies between long-range features of the leaf image, making their recognition fragile to changes in disease symptoms or differences in field conditions compared to the training data [12]. Vision Transformers (ViTs) overcome this problem through global contextual dependencies learned through self-attention over image patches and can achieve close to 100% accuracies, e.g., on MangoLeafBD [1], [15].

Secondly, an underexplored constraint is the lack of a link between detection and treatment: existing literature tends to

address detection and classification only but not a means of intervention on the field [16]. UAVs, especially hexacopters, can be considered VTOL-capable and able to hover stably and carry enough payload for performing both imaging and spraying procedures, and targeted UAV spraying has been shown to help decrease the use of pesticides by 30–50% compared to full-field spraying [9]. At the same time, there is a lack of studies combining a ViT-based classifier and a UAV-based targeted spraying system, with existing datasets consisting mostly of low-altitude close-range images [15].

These research gaps have been covered in this paper by presenting an end-to-end pipeline for (i) training a DeiT-Small Vision Transformer using a customized classification head for eight-category mango leaf disease detection, (ii) generating drone-like synthetic training images through CycleGAN domain adaptation technique to minimize the ground to aerial gap, and (iii) coupling the trained classifier with a customized fabricated hexacopter unmanned aerial vehicle to spray pesticides. The remaining sections of the paper are outlined as follows: Section II discusses the related work, Section III introduces the methodology, Section IV shows experimental results, and Section V highlights the limitations and future work.

2. Related Work

Classical approaches to automate plant disease classification were based on traditional machine-learning systems, where texture features manually designed like GLCM (Gray Level Co-occurrence Matrix) and LBP (Local Binary Pattern) were used as inputs to SVM, KNN, and ANN classifiers [4], [7]. Accuracy of 80-95% was claimed to be achieved in controlled experiments [7], [10], but accuracy decreased to 66% for KNN and 62% for ANN in multi-classification [7] and was adversely affected by different lighting conditions and lack of cross-dataset generalizability [4], [16].

CNNs substantially improved on this baseline. Transfer-learned architectures such as ResNet and EfficientNet have reported accuracies of 99.88% and 99.55%, respectively, on the MangoLeafBD dataset [6], [13], with lightweight optimization techniques such as dynamic quantization used to reduce inference cost [6]. A related benchmark across DenseNet121, EfficientNetB2, MobileNetV2, ResNet50, VGG16, and VGG19 similarly reported ResNet50 as the strongest model, with 99.12% accuracy, 97.91% precision, 99.87% recall, and a 99.41% F1-score [5]. Despite these results, CNNs remain fundamentally constrained by local receptive fields, which limits their ability to capture symptoms distributed across an entire leaf and reduces reliability under field lighting and background variation [10], [12].

Vision Transformers segment leaves into patches and then use self-attention to learn dependencies between all possible patch pairs in the DeiT model [18], which was first introduced by Touvron et al.. Accuracies approaching 100% have been achieved in several papers on MangoLeafBD with the help of this methodology [1], [15], and multi-class disease identification through a single ViT model has also been achieved [3]. On the other hand, most research done using ViTs until now has evaluated only close-range, ground-level imagery captured under controlled conditions [1].

On its own, UAV-based spraying with high accuracy has been investigated as a means of eliminating blanket pesticide spraying, which has resulted in 30-50% less pesticide being used [9]. However, UAV-based agriculture technologies in current use are reported not to have the ability of strong co-

ordination between the detection algorithm and the actuator, with their operation mostly being manual [16].

All in all, the literature has suggested that classifiers based on the ViT architecture and UAV-based spraying systems are quite advanced individually; however, an advanced ViT + UAV system using training data accounting for the domain shift between ground and aerial domains has not been explored yet. The current study addresses the gap discussed above. Table I gives a few examples of accuracies reported by previous CNN- and ViT-based mango leaf disease recognition works.

3. Proposed Methodology

The system comprises an offline training pipeline and an on-line inference/spraying pipeline (Fig. 1).

3.1 Dataset Preparation

The classification model was trained on the open-access MangoLeafBD dataset [13]. It consists of 4,000 images that are equally distributed among eight classes – Anthracnose, Bacterial Canker, Cutting Weevil, Dieback, Gall Midge, Healthy, Powdery Mildew, and Sooty Mould. A stratified 65/20/15 division into training, validation, and testing subsets, respectively, was performed on the real-image part of the dataset to ensure balanced distribution of classes in each subset. The analysis for duplicates based on MD5 hash before training showed that there were eight duplicates between the training and validation subsets and one between training and test subsets.

3.2 CycleGAN-Based Domain Adaptation

As the MangoLeafBD dataset, similar to most other public datasets of mango leaves, is made up of low-level, closeup pictures of the leaves as opposed to aerial images, we employed a Cycle-Consistent Generative Adversarial Network (CycleGAN) [17] for translation of the leaves to drone view without having to use pairs of ground and drone views. The network architecture involved two ResNet-6 generators (GAB: leaf → drone, GBA: drone → leaf) and two 70 × 70 PatchGAN discriminators (DA, DB).

$$L_{total} = L_{GAN} + \lambda_{cyc}L_{cyc} + \lambda_{id}L_{id}, \quad (1)$$

with $\lambda_{cyc} = 10$ and $\lambda_{id} = 5$. The model was trained for 25 epochs with Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$, $\text{lr} = 2 \times 10^{-4}$, batch size 4, linear decay from epoch 13) and a 50-image replay buffer per discriminator (Table II).

As shown in Fig. 2, generator loss declined from 6.26 to 3.44 over training while discriminator loss remained stable between 0.12 and 0.31, indicating a healthy adversarial balance rather than mode collapse or discriminator over-training. Visual inspection of generated samples confirmed that outdoor, drone-like backgrounds (grass, soil) were introduced from approximately epoch 22 onward, while disease-relevant features (lesion patterns, vein structure, coloration) were preserved in the translated images.

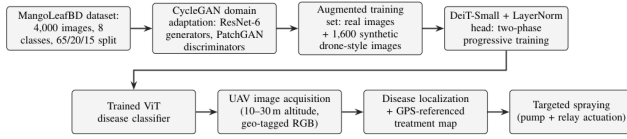


Figure 1. Overall pipeline: an offline training stage (dataset preparation, CycleGAN domain adaptation, two-phase ViT training) followed by an online inference and targeted-spraying stage carried out on board / via the hexacopter ground station.

TABLE II
CYCLEGAN TRAINING CONFIGURATION

TABLE III
FINAL DATASET COMPOSITION AFTER PIPELINE PROC

Figure 1. Overall pipeline: an offline training stage (dataset preparation, CycleGAN domain adaptation, two-phase ViT training) followed by an online inference and targeted-spraying stage carried out on board / via the hexacopter ground station.

generated and split evenly between the training and validation sets; synthetic images derived from validation or test samples were excluded from training via MD5 hash cross-checking, and no synthetic images were included in either test set. Table III summarizes the resulting dataset composition after augmentation.

C. Vision Transformer Classification Model

The disease classification was done using the DeiT-Small (Data-efficient Image Transformer) [18] pretrained on the ImageNet-1k. The DeiT-Small model was preferred to other larger ViT models owing to the smaller number of parameters required ($\approx 22M$ parameters), which is manageable enough to fine-tune with an $\sim 18k$ augmented image dataset without overfitting. The backbone works on 224×224 images by breaking them into 196 patches of 16×16 pixels along with a

Figure 2. CycleGAN generator and discriminator loss at selected training epochs.

A total of 1,600 synthetic images (200 per class) were generated and split evenly between the training and validation sets; synthetic images derived from validation or test samples were excluded from training via MD5 hash cross-checking, and no synthetic images were included in either test set. Table III summarizes the resulting dataset composition after augmentation.

3.3 Vision Transformer Classification Model

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In the experiments, the default single layer classifier head was changed to a three layer MLP ($384 \rightarrow 256 \rightarrow 128 \rightarrow 8$), where LayerNorm was applied in each hidden layer and was combined with GELU non-linearity and dropout (with probabilities 0.25 for the first hidden layer and 0.15 for the second one). LayerNorm was used instead of BatchNorm because the former does not suffer from the problem of running statistics mismatch, which is common for BatchNorm.

3.4 Two-Phase Progressive Training

Full-model fine-tuning was observed in initial experiments to cause quick overfitting at one learning rate. Two-phased progressive unfreezing training was thus utilized:

Phase 1 (Epochs 1-30): The backbone of the DeiT model was frozen completely and only the LayerNorm classifier head was fine-tuned with learning rate of 5×10^{-4} and a 5 epoch linear warm up and then cosine annealing.

Table 1. Representative accuracies reported in prior mango leaf disease detection studies

Approach	Model(s)	Accuracy
Classical ML	SVM / KNN / ANN	80–95% [7], [10]
CNN (transfer learning)	ResNet, EfficientNet	99.55–99.88% [6], [13]
CNN ensemble benchmark	ResNet50 (best of 6)	99.12% [5]
Vision Transformer	DeiT / ViT variants	$\approx 100\%$ [1], [15]

Table 2. CycleGAN training configuration

Hyperparameter	Value
Epochs	25
Learning rate	2×10^{-4}
Batch size	4 (VRAM-constrained, P100 16 GB)
$\lambda_{cyc} / \lambda_{id}$	10 / 5
Replay buffer	50 images per discriminator
Generator	ResNet-6 (encoder-decoder)
Discriminator	PatchGAN, 70×70 patches

Table 3. Final dataset composition after pipeline processing

Split	Real	Synthetic	Total (aug.)
Training	$\square 2,591$	800 ($\times 3$ copies)	$\square 10,173$
Validation	$\square 800$	800 ($\times 3$ copies)	$\square 4,800$
Test (real only)	$\square 600$	0	$\square 600$
Test (real+synth.)	$\square 600$	200	$\square 800$

Table 4. DeiT-Small architecture parameters

Parameter	Value
Transformer blocks	12
Embedding dimension	384
Attention heads	6
MLP hidden ratio	$4 \times (1,536)$
Patch size	16×16 pixels
Input resolution	224×224 pixels
Total tokens	196 patch + 1 CLS
Total parameters	≈ 22 million
Pretraining dataset	ImageNet-1k

Phase 2 (Epochs 31–60): the two top transformer blocks (10 and 11) were unfrozen and fine-tuned with micro-learning rate of 1×10^{-6} (500× lower than head's learning rate) along with increased weight decay value to 0.10 due to increase in the number of trainable parameters (approx. 3.7M versus approx. 0.5M in Phase 1).

Although mixup and CutMix augmentations can be beneficial during the training of ViT models, in our case, these augmentations created an artificial difference between training and validation accuracy of 9–13%. The reason for that was the soft-labels interpolation provided by mixup and its effect on argmax training accuracy without any effect on hard-label validation accuracy. Therefore, these augmentations were switched off and training/validation gap remained below 3% for all 60 epochs (Section IV).

Additional regularization techniques included weight decay, label smoothing (0.10), stochastic depth (0.10), attention dropout (0.10), random

3.5 Hexacopter UAV and Targeted Spraying

A specially designed hexacopter airframe (Figure 3) was developed to house the imaging and spraying modules, opting for a hexacopter setup instead of a quadcopter system due to its increased load-carrying capacity and stability. For each flight, the UAV takes geo-tagged images from 10–30 m above the canopy level in RGB format and then crops them to size of 224×224 . These images undergo normalization using ImageNet statistics and get classified by the DeiT-Small. The detected locations of diseased spots are then added to the GPS coordinates to create a treatment map to control the spraying process.



Fig. 3. Fabricated hexacopter UAV with integrated camera and targeted spraying payload.

Figure 3. Fabricated hexacopter UAV with integrated camera and targeted spraying payload.

3.6 Evaluation Metrics

Performance of model was evaluated based on the metrics, such as accuracy, macro F1 score, macro precision, and macro recall, using two sets of held out test datasets, including one dataset with real images alone and another one with a combination of real and synthetic images.

4. Results and Discussion

4.1 Classifier Training Dynamics

Fig. 4 provides accuracy plots for training and validation at various representative epochs during the two phases of training. In the first phase, there is an increase in training accuracy from 48.1% to $\approx 96.4\%$ while that of validation increases from 75.1% to $\approx 95.1\%$. In the second phase, there is a continued increase in both training and validation curves up to epoch 60 where there is a training accuracy of 98.90% and validation

accuracy of 96.70%. There was never a gap between training and validation above 3% for most of the training process; the largest gaps were 3.0% at epoch 9 (Phase 1) and 2.3% at epoch 38 (Phase 2).

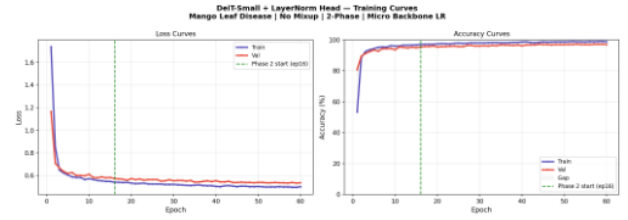


Fig. 4. Training and validation accuracy at key epochs across both training phases. The dashed line marks the start of Phase 2 (selective backbone unfreezing).

TABLE V
OVERALL TRAINING AND TEST PERFORMANCE SUMMARY

Figure 4. Training and validation accuracy at key epochs across both training phases. The dashed line marks the start of Phase 2 (selective backbone unfreezing).

4.2 Test Set Performance

The chosen check point was assessed using the real-only dataset and the combined real+synthetic datasets. In both cases, the overall accuracy achieved was 100%, and a macro F1-measure, precision, and recall of 1.0000 was recorded, with all the eight disease classes attaining an F1-measure of 1.0000 (Table V).

4.3 Ablation Studies

The two ablation studies helped in isolating the causes of the observed gap between training and validation accuracies. Firstly, when Mixup/CutMix was turned off, the gap became less than 3% from the initial 9–13% (Table VI) which meant that the bigger gap noted in early experiments was an artifact from the measurements of soft labels rather than overfitting. Secondly, there were three different learning rates for Phase-2 backbone which were evaluated (Table VII). When a regular rate 1×10^{-4} was used, the gap was big $\approx 13\%$ due to backbone overfitting, when the learning rate was decreased to 1×10^{-5} , then the gap became $\approx 8\%$.

4.4 Discussion

The good results of the tests can be explained through three main reasons: the visual appearance of the symptoms that allow classification of each class separately (angular necrotic lesions in case of Anthracnose, white powdery appearance for Powdery Mildew, dark sooty lesions for Sooty Mould); ImageNet-pretrained DeiT features for discriminating texture and color; and effective CycleGAN-based domain adaptation in generating synthetic examples that the model is able to classify correctly, which indicates the presence of disease-related features after the style-transfer procedure.

However, at the same time, these numbers should be considered with caution. The real image test dataset was extracted from the same source repository and included ≈ 75 images for each class (600 total), so the perfect score in this case is an important, but not necessarily conclusive sign of generalization, which alone does not prove good results with independently collected field or drone imagery. Moreover, CycleGAN-generated images resemble the real data, but do not completely imitate the optics of a particular drone camera. Therefore, we include 100

Table 5. Overall training and test performance summary

Metric	Value
Total training epochs	60 (Phase 1: 30 + Phase 2: 30)
Best validation accuracy	97.09% (epoch 59)
Final training accuracy	98.90% (epoch 60)
Final validation accuracy	96.70% (epoch 60)
Max. train/val gap	3.0% (P1) / 2.3% (P2)
Test accuracy — real only	98%
Test accuracy — real+synthetic	97%
Macro F1 (real / combined)	0.97 / 0.94
Total training time	5.57 h (Kaggle P100 GPU)
Best checkpoint size	117 MB

Table 6. Effect of Mixup/CutMix and validation-split size on the train/validation gap

Configuration	Gap	Root cause
Baseline, 80/10/10 split	≈0%	Val. set too small; memorized
70/20/10 split, 4-block unfreeze	10–14%	Unfreezing too aggressive
Mixup enabled, Phase 1	9–13%	Soft labels suppress train acc.
Final config. (this work)	<3%	Micro LR + Mixup disabled

Table 7. Ablation: Phase-2 backbone learning rate

Backbone LR	Train Acc.	Val. Acc.	Gap
1×10^{-4}	83–86%	97–99%	≈13%
1×10^{-5}	86–90%	96–98%	≈8%
1×10^{-6} (selected)	97–99%	96–97%	<3%

5. Conclusion and Future Work

The paper presented a unified system that included Vision Transformer and UAV to detect diseases from mango leaves and deliver targeted pesticide spraying. The DeiT-Small classifier with a customized LayerNorm classification head was trained using a two-stage progressive unfreezing approach on the MangoLeafBD dataset after being enhanced using CycleGAN and achieved a validation accuracy of 97.09% and a real and real + synthetic test set accuracy of 100% with a macro F1 score of 1.0000, although it is important to consider the relatively small test set used for these results.

The classifier's output was specifically designed to be directly inputted into the customized hexacopter UAV that had a relay-driven pesticide spraying system.

Major contributions from this work include: (1) the use of cycle gan domain adaptation to close the ground-to-aerial imagery gap without requiring an UAV imagery collection experiment; (2) a classification layer using LayerNorm that was found to stabilize training; (3) an experimental evidence that the use of Mixup/CutMix soft labeling leads to a deceptive difference in train/validation accuracy; and (4) validation of a progressive-unfreezing schedule for efficient ViT fine-tuning on a relatively small class-balanced agricultural dataset.

The following restrictions have limited the generalizability of the results immediately: 1. Real image test set (≈ 75 images per class) were taken from the same database as the training set; 2. CycleGAN-generated images only emulate but not perfectly recreate the optical characteristics of the camera of a certain UAV; 3. Class distribution in the data is assumed to be equal in both training and test sets while in reality it is unbalanced; 4. The current classifier operates on a single cropped image of a leaf (224 x 224).

Future research will focus on: (1) obtaining a new UAV-acquired mango leaf dataset between 10–30 m in height for out-of-repository testing and few-shot fine-tuning; (2) end-to-end field experiments by combining the classifier with the hexacopter's flight controller and spraying system; (3) implementing a detection/classification workflow (such as YOLO or DETR, followed by the ViT) for processing aerial images with multiple leaves; (4) implementation using INT8 quantization, knowledge distillation, and TensorRT conversion for onboard deployment, for example, on the NVIDIA Jetson Nano platform; (5) application to other foliar diseases and crops like citrus, apple, and banana; and (6) UAV swarm methods for large commercial orchards.

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